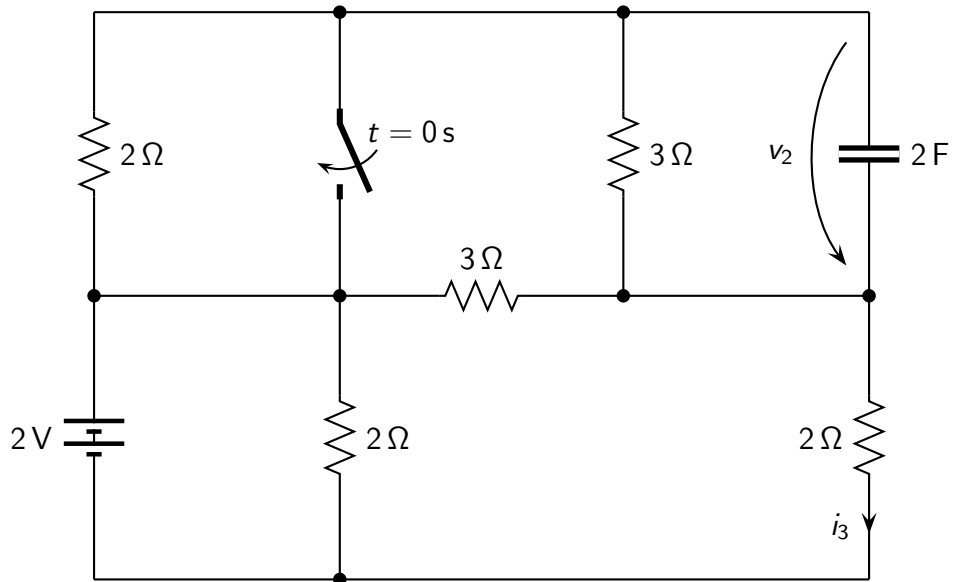


Problem: find the expression of $v_2(t)$, $i_3(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{12}{7} \text{ s}$;
- Initial conditions before switching: $v_2(0^-) = \frac{-18}{31} \text{ V}$;
- Initial conditions after switching: $v_2(0^+) = \frac{-18}{31} \text{ V}$; $i_3(0^+) = \frac{22}{31} \text{ A}$;
- Steady-state solution: $v_2(\infty) = \frac{-6}{7} \text{ V}$; $i_3(\infty) = \frac{4}{7} \text{ A}$;
- Solution for $t > 0$:

$$v_2(t) = \left(0.276 e^{-t/\tau} + \frac{-6}{7} \right) \text{ V}$$

$$i_3(t) = \left(0.138 e^{-t/\tau} + \frac{4}{7} \right) \text{ A}$$